Same Strong Top Prediction, Different Confidence

Illustrative example: "PO" has probability 0.60 in both cases. The warning is probability escaping behind it.

TOP-5 the five highest-probability next tokens → **m_5** their probability mass added together → **TAIL** every token ranked 6 through |V|

A · Mostly harmless case variation

The head is filled with plausible forms such as "PO" and "po".

Top-1: P("PO") = 0.60

Five most likely next-token candidates

PO	po	Po	P	pO
0.60	0.22	0.07	0.05	0.03

Where the full 100% probability mass goes



top-5 retained 97% tail 3%

Only 3% escapes → higher confidence

B · Strong top-1, but probability leaks

The top token is unchanged, yet much more mass moves outside the candidate set.

Top-1: P("PO") = 0.60

Five most likely next-token candidates

PO	po	Po	P	pO
0.60	0.08	0.04	0.03	0.03

Where the full 100% probability mass goes



top-5 retained 78% tail 22%

22% escapes → lower confidence

Why k = 5? k=2 .741 | k=5 .749 | k=10 .750

AUROC: k=5 is the smallest cutoff on the performance plateau—not a mathematical constant.